

Challenge no. 4 on AWS

Use Case: Setting up Transit Gateway and VPC Endpoints for a Multi-VPC Architecture

Scenario:

A large organization is migrating its on-premises infrastructure to the AWS cloud.

The organization's architecture involves multiple VPCs for different departments and applications, each requiring secure communication with centralized services and external resources.

The IT team needs to design and implement a scalable and efficient network architecture to accommodate the organization's growth and ensure robust connectivity between VPCs and external services.

Objectives:

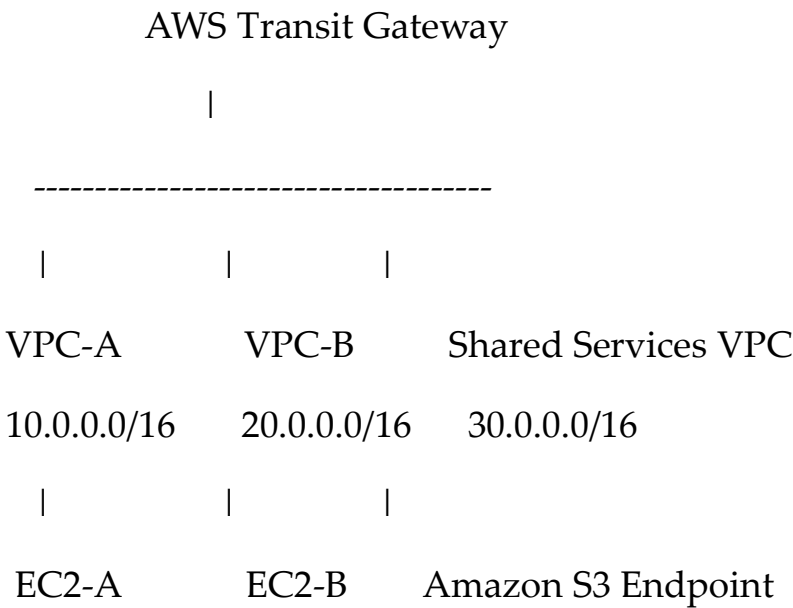
- Design and deploy a scalable network architecture using AWS Transit Gateway to simplify network connectivity between multiple VPCs.
- Configure VPC endpoints to securely access AWS services without internet gateways or NAT gateways, ensuring data privacy and minimizing exposure to external threats.

Objective: Create multiple VPCs.

- Connect the VPCs using AWS Transit Gateway.
- Configure routing between VPCs.
- Create a VPC Endpoint to securely access Amazon S3 without using an Internet Gateway or NAT Gateway.
- Verify connectivity.

Prerequisites

- AWS Account
- IAM User with Administrator Access
- Basic knowledge of AWS VPC
- AWS Region (Example: **US East (N. Virginia) - us-east-1**)



Step 1: Login to AWS Console

- Login to AWS Management Console.
 - Select Region **US East (N. Virginia)**.
 - Search **VPC**.
 - Open the VPC Dashboard.
-

Step 2: Create VPC-A

Go to

VPC → Your VPCs → Create VPC

Enter

Setting	Value
Name	VPC-A
IPv4 CIDR	10.0.0.0/16
IPv6	None
Tenancy	Default

Click

Create VPC

Enable DNS Hostname

Select **VPC-A**

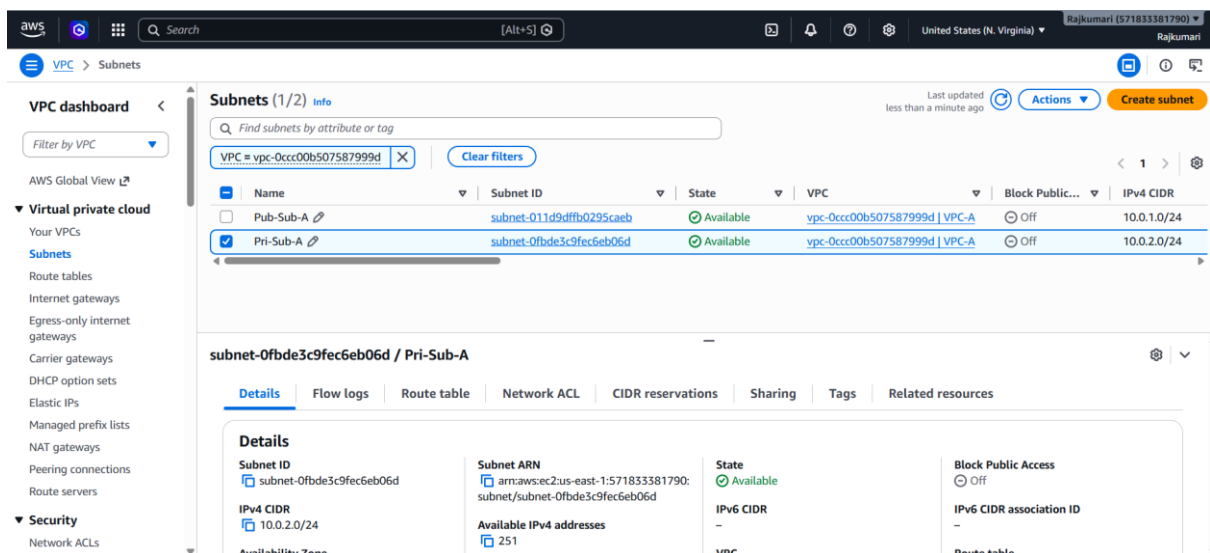
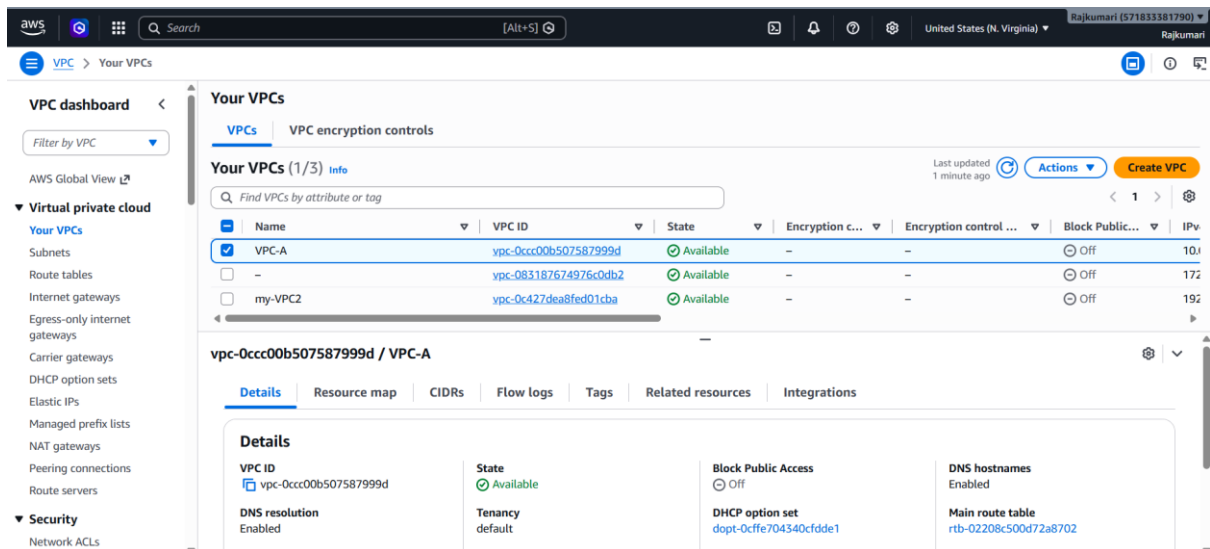
Actions

Edit VPC Settings

Enable

- DNS Resolution
- DNS Hostnames

Save.



VPC → Internet Gateways

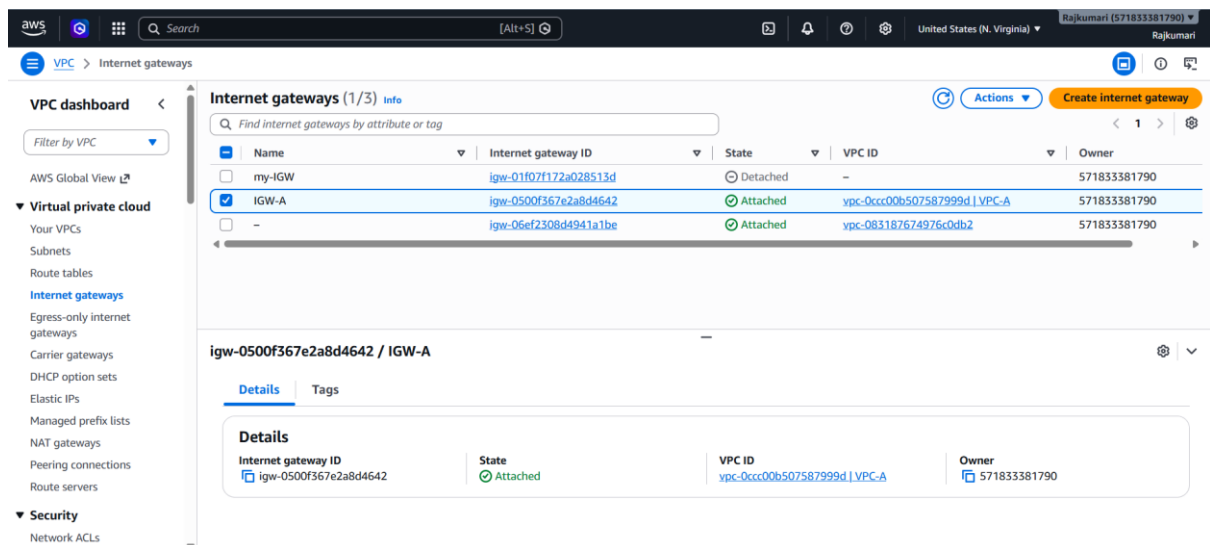
Create Internet Gateway

Name

IGW-A

Attach to

VPC-A



Step 5: Configure Route Table

Create

Public Route Table

Add routes

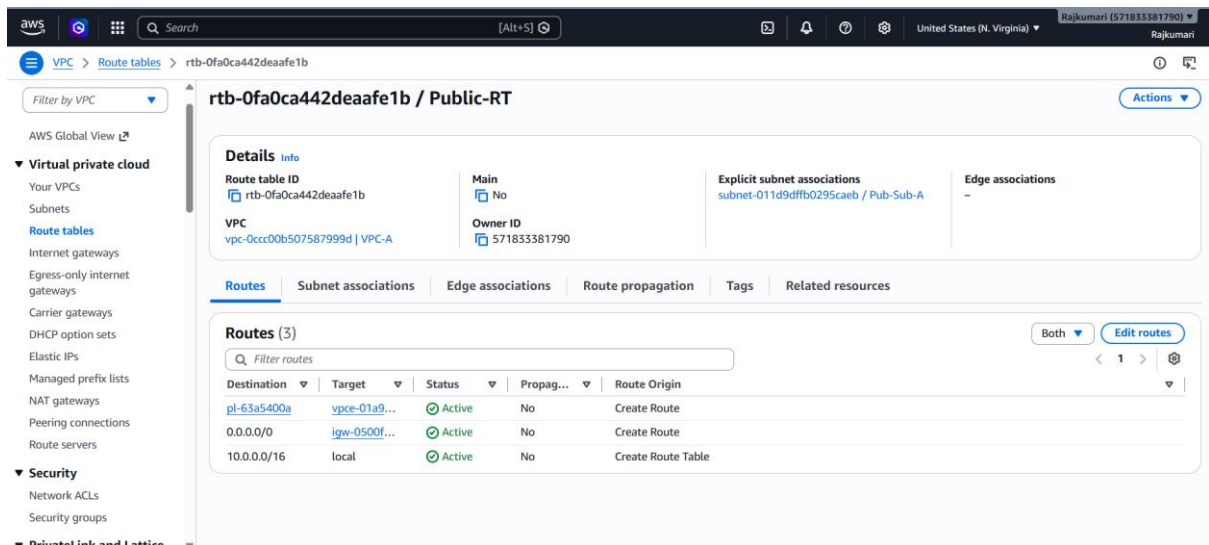
Destination Target

10.0.0.0/16 Local

0.0.0.0/0 Internet Gateway

Associate with

Public Subnet.



Step 6: Launch EC2 Instance in VPC-A

Launch Instance

Setting	Value
Name	EC2-A
AMI	Amazon Linux 2023
Instance Type	t2.micro
Key Pair	Existing PEM
Network	VPC-A
Subnet	Public-Subnet-A
Auto Assign Public IP	Enable
Security Group	Allow
SSH	(22)

The screenshot displays the AWS Management Console interface. The top navigation bar shows the user is logged in as 'Rajkumari' in the 'United States (N. Virginia)' region. The main content area is titled 'Launch an instance' and shows a green success banner: 'Success: Successfully initiated launch of instance (i-045730f96d46e6b81)'. Below this, a 'Launch log' section lists the steps: 'Initializing requests', 'Creating security groups', 'Creating security group rules', and 'Launch initiation', all marked as 'Succeeded'. A 'Next Steps' section provides suggestions for further actions like 'create alarm' or 'create backup'. Four task cards are visible: 'Create billing usage alerts', 'Connect to your instance', 'Connect an RDS database', and 'Create EBS snapshot policy'. The bottom section shows the 'Instances' list with one instance, 'EC2-A', in the 'Running' state. The instance details include its ID, type (t3.micro), status (Initializing), availability zone (us-east-1a), and public IP address.

Success
Successfully initiated launch of instance (i-045730f96d46e6b81)

Launch log

- Initializing requests: Succeeded
- Creating security groups: Succeeded
- Creating security group rules: Succeeded
- Launch initiation: Succeeded

Next Steps
What would you like to do next with this instance, for example "create alarm" or "create backup"

Create billing usage alerts
To manage costs and avoid surprise bills, set up email notifications for billing usage thresholds.
[Create billing alerts](#)

Connect to your instance
Once your instance is running, log into it from your local computer.
[Connect to instance](#)
[Learn more](#)

Connect an RDS database
Configure the connection between an EC2 instance and a database to allow traffic flow between them.
[Connect an RDS database](#)
[Create a new RDS database](#)
[Learn more](#)

Create EBS snapshot policy
Create a policy that automates the creation, retention, and deletion of EBS snapshots
[Create EBS snapshot policy](#)

Instances (1)

Instance state: Running
Find Instance by attribute or tag (case-sensitive)
Instance state = running
Clear filters

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	Put
EC2-A	i-045730f96d46e6b81	Running	t3.micro	Initializing	us-east-1a	ec2-44-198-175-187.co...	44...

Select an instance

Step 7: Create VPC-B

Repeat the same process.

Setting Value

Name VPC-B

CIDR 20.0.0.0/16

Create

- Public Subnet
- Private Subnet
- Internet Gateway

- Route Table
- EC2 Instance

Virtual private cloud

Your VPCs (1/3)

Name	VPC ID	State	Encryption c...	Encryption control ...	Block Public...	IPv
VPC-A	vpc-0ccc00b507587999d	Available	-	-	Off	10.0.1.0/24
-	vpc-083187674976c0db2	Available	-	-	Off	172.16.0.0/16
VPC-B	vpc-02bc6afdf8e0871c1	Available	-	-	Off	20.0.1.0/24

vpc-02bc6afdf8e0871c1 / VPC-B

Details

VPC ID vpc-02bc6afdf8e0871c1	State Available	Block Public Access Off	DNS hostnames Disabled
DNS resolution Enabled	Tenancy default	DHCP option set dopt-0cffe704340cfdde1	Main route table -

- Public Subnet
- Private Subnet

Subnets (4)

Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR
Pub-Sub-B	subnet-0511da5d048bda8f3	Available	vpc-02bc6afdf8e0871c1 VPC-B	Off	20.0.1.0/24
Pub-Sub-A	subnet-011d9dfb0295cae8	Available	vpc-0ccc00b507587999d VPC-A	Off	10.0.1.0/24
Pri-Sub-B	subnet-0192a73d68389cfba	Available	vpc-02bc6afdf8e0871c1 VPC-B	Off	20.0.2.0/24
Pri-Sub-A	subnet-0fbde3c9fec6eb06d	Available	vpc-0ccc00b507587999d VPC-A	Off	10.0.2.0/24

Select a subnet

- Internet Gateway

VPC dashboard < **igw-0c4a6a0ef5cb37276 / IGW-B** Actions

Details [Info](#)

Internet gateway ID: [igw-0c4a6a0ef5cb37276](#) | State: Attached | VPC ID: [vpc-02bc6afdf8e0871c1 | VPC-B](#) | Owner: [571833381790](#)

Tags (1) Manage tags

Search tags

Key	Value
Name	IGW-B

- Route Table

Route tables Updated routes for rtb-0f748089e6c7c4b4c / Public-RT-B successfully Details

Route tables (1/6) [Info](#) Last updated less than a minute ago Actions Create route table

Find route tables by attribute or tag

VPC = [vpc-0ccc00b507587999d](#) X VPC = [vpc-02bc6afdf8e0871c1](#) X Clear filters

	Name	Route table ID	Explicit subnet associations	Edge associations	Main	VPC
☐	Public-RT-A	rtb-0fa0ca442deaaf1b	subnet-011d9dffb0295cae / Pub-Sub-A	-	No	vpc-0ccc00b507587999d
☐	-	rtb-01f35982854beaf3d	-	-	Yes	vpc-02bc6afdf8e0871c1
☐	-	rtb-02208c500d72a8702	-	-	Yes	vpc-0ccc00b507587999d
☐	Private-RT-A	rtb-0fb822525e15963fe	subnet-0fbde3c9fec6eb06d / Pri-Sub-A	-	No	vpc-0ccc00b507587999d
☒	Public-RT-B	rtb-0f748089e6c7c4b4c	-	-	No	vpc-02bc6afdf8e0871c1
☐	Private-RT-B	rtb-0e59dccc0de557460	-	-	No	vpc-02bc6afdf8e0871c1

rtb-0f748089e6c7c4b4c / Public-RT-B Details Routes Subnet associations Edge associations Route propagation Tags Related resources

EC2 Instance

EC2 > Instances > Launch an instance

Success
Successfully initiated launch of instance (i-050f5f20582d24983)

Launch log

Initializing requests	✓ Succeeded
Creating security groups	✓ Succeeded
Creating security group rules	✓ Succeeded
Launch initiation	✓ Succeeded

Next Steps
What would you like to do next with this instance, for example "create alarm" or "create backup"

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[Create EBS snapshot policy](#)

EC2 > Instances

Instances (2) Info
Last updated less than a minute ago

Connect Instance state Actions Launch instances

Saved filter sets
Choose filter set Find Instance by attribute or tag (case-sensitive)

Instance state = running Clear filters

	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	Put
☐	EC2-A	i-045730f96d46e6b81	Running	t3.micro	3/3 checks passed	us-east-1a	ec2-44-198-175-187.co...	44
☐	EC2-B	i-050f5f20582d24983	Running	t3.micro	Initializing	us-east-1b	-	-

Select an instance

EC2
Dashboard
AWS Global View
Events
▼ **Instances**
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
Capacity Manager
▼ **Images**
AMIs
AMI Catalog
▼ **Elastic Block Store**
Volumes
Snapshots
Lifecycle Manager
▼ **Network & Security**
Security Groups

Step 8: Create Shared Services VPC

Create another VPC

Setting Value

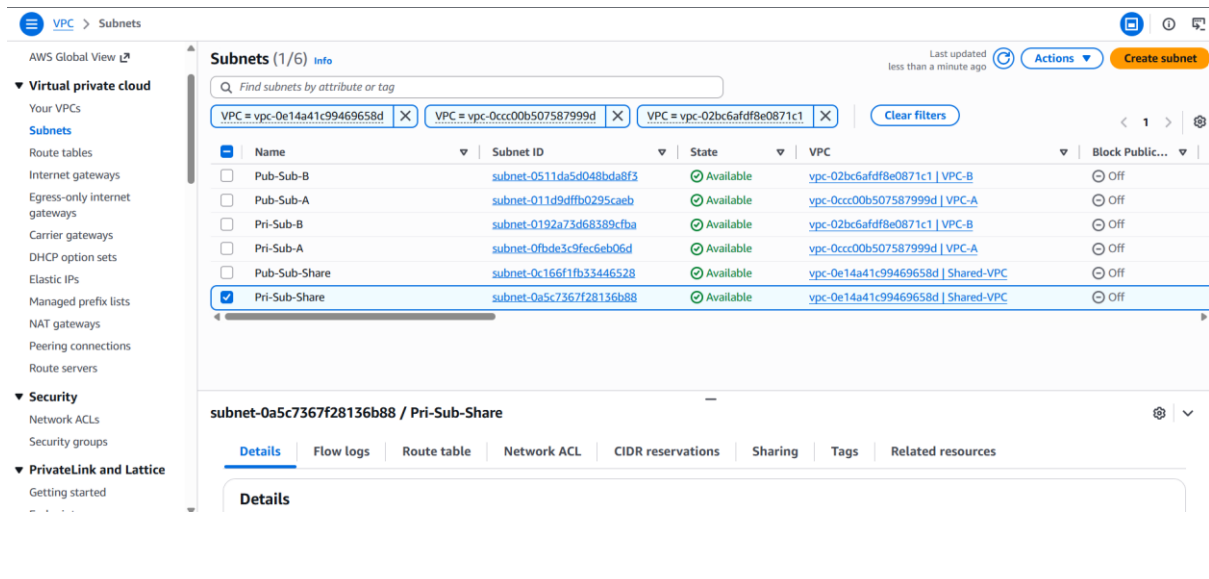
Name Shared-VPC

CIDR 30.0.0.0/16

Create

- Public Subnet
- Private Subnet

Launch one EC2 instance if required.



Step 9: Create Transit Gateway

Go to

VPC

Transit Gateways

Click

Create Transit Gateway

Enter

Setting

Value

Name

Organization-TGW

ASN

Default

DNS Support

Enable

VPN ECMP

Enable

Auto Accept Shared Attachments Disable

Click

Create.

Wait until

Available

Summary Of Resources

VPC-A (10.0.0.0/16)

└— **IGW-A**

└— **Public-RT-A**

└— **Private-RT-A**

VPC-B (20.0.0.0/16)

└— **IGW-B**

└— **Public-RT-B**

└— **Private-RT-B**

Shared-VPC (30.0.0.0/16)

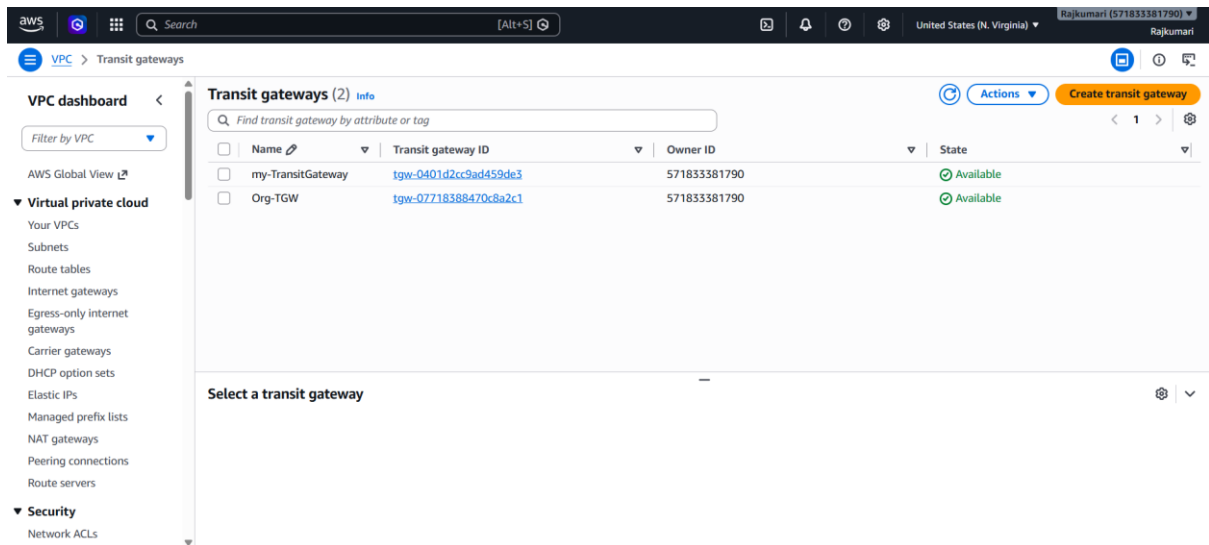
└— **IGW-Shared**

└— **Public-RT-Shared**

└— **Private-RT-Shared**



AWS Transit Gateway



Step 10: Attach VPC-A to Transit Gateway

Go to

Transit Gateway Attachments

Create Attachment

Choose

Setting	Value
---------	-------

Resource Type	VPC
---------------	-----

Transit Gateway	Organization-TGW
-----------------	------------------

VPC	VPC-A
-----	-------

Subnet	Private-Subnet-A
--------	------------------

Create Attachment.

Wait

Available.

☰ VPC > Transit gateway attachments > Create transit gateway attachment ⓘ 🖨

VPC attachment

Select and configure your VPC attachment.

☒ DNS support ⓘ

☒ Security Group Referencing support ⓘ

☐ IPv6 support ⓘ

☐ Appliance Mode support ⓘ

VPC ID

Select the VPC to attach to the transit gateway.

vpc-0ccc00b507587999d ▼

Subnet IDs ⓘ

Select the subnets in which to create the transit gateway VPC attachment.

☒ us-east-1a subnet-0fbde3c9fec6eb06d ▼

☐ us-east-1b No subnet available

☐ us-east-1c No subnet available

☐ us-east-1d No subnet available

☐ us-east-1e No subnet available

☐ us-east-1f No subnet available

subnet-0fbde3c9fec6eb06d ✕

Step 11: Attach VPC-B

Repeat

Choose

VPC-B

Private Subnet

☰ VPC > Transit gateway attachments > Create transit gateway attachment ⓘ 🖨

VPC attachment

Select and configure your VPC attachment.

☒ DNS support ⓘ

☒ Security Group Referencing support ⓘ

☐ IPv6 support ⓘ

☐ Appliance Mode support ⓘ

VPC ID

Select the VPC to attach to the transit gateway.

vpc-02bc6afdf8e0871c1 ▼

Subnet IDs ⓘ

Select the subnets in which to create the transit gateway VPC attachment.

☐ us-east-1a No subnet available

☒ us-east-1b subnet-0192a73d68389cfba ▼

☐ us-east-1c No subnet available

☐ us-east-1d No subnet available

☐ us-east-1e No subnet available

☐ us-east-1f No subnet available

subnet-0192a73d68389cfba ✕

Step 12: Attach Shared Services VPC

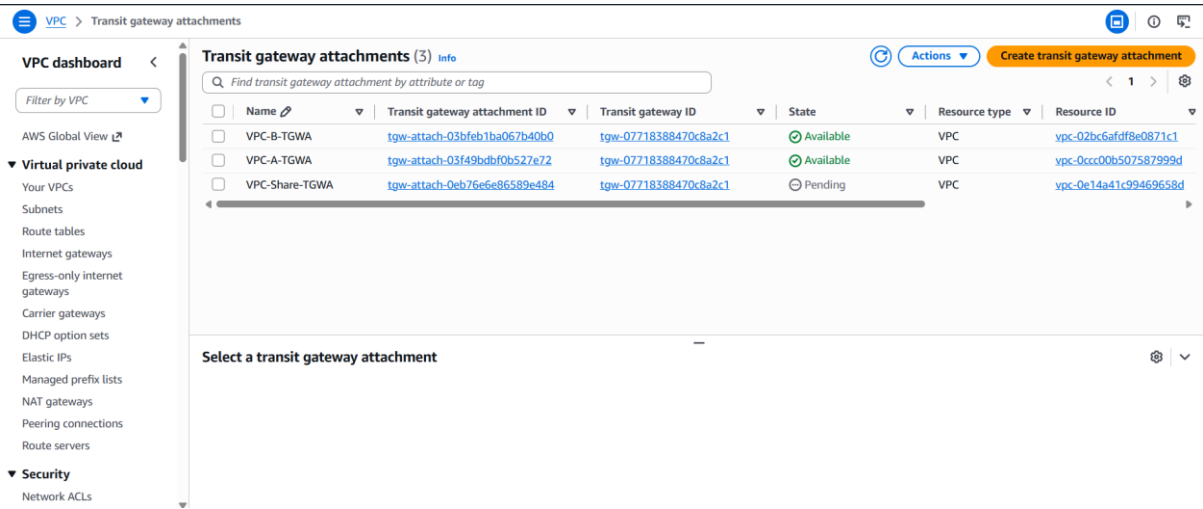
Repeat

Choose

Shared-VPC

Private Subnet

Create Attachment.



Step 13: Update Private Route Tables

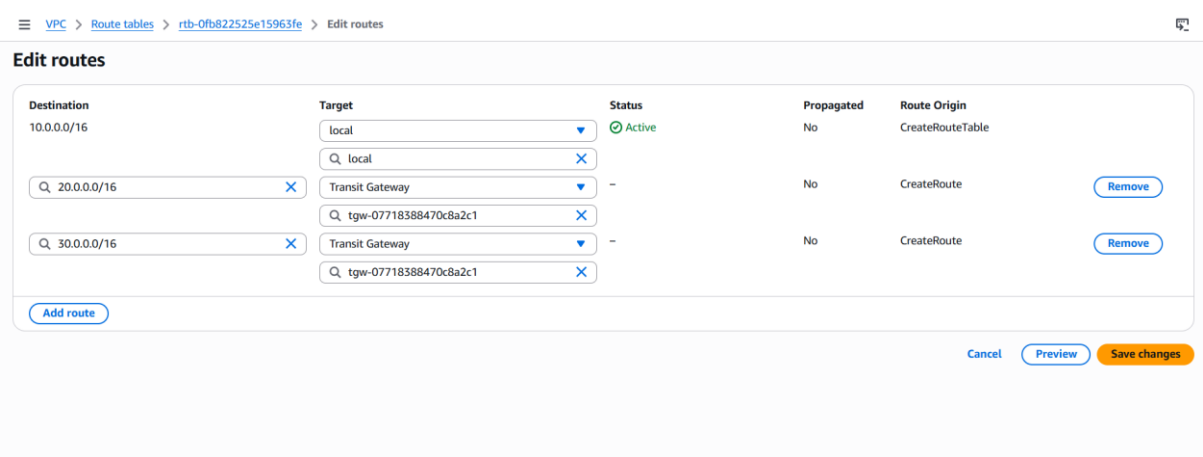
VPC-A Route Table

Add

Destination Target

20.0.0.0/16 Transit Gateway

30.0.0.0/16 Transit Gateway



VPC-B Route Table

Add

Destination Target

10.0.0.0/16 Transit Gateway

30.0.0.0/16 Transit Gateway

VPC > Route tables > rtb-0e59dccc0de557460 > Edit routes

Edit routes

Destination	Target	Status	Propagated	Route Origin	
20.0.0.0/16	local	Active	No	CreateRouteTable	
10.0.0.0/16	local				
	Transit Gateway	Active	No	CreateRoute	Remove
30.0.0.0/16	tgw-07718388470c8a2c1				
	Transit Gateway	Active	No	CreateRoute	Remove
tgw-07718388470c8a2c1					

Add route

Cancel

Preview

Save changes

Shared Services VPC Route Table

Add

Destination Target

10.0.0.0/16 Transit Gateway

20.0.0.0/16 Transit Gateway

VPC > Route tables > rtb-0c5774ba902dae6a9 > Edit routes

Edit routes

Destination	Target	Status	Propagated	Route Origin	
30.0.0.0/16	local	Active	No	CreateRouteTable	
10.0.0.0/16	local				
	Transit Gateway	Active	No	CreateRoute	Remove
20.0.0.0/16	tgw-07718388470c8a2c1				
	Transit Gateway	Active	No	CreateRoute	Remove
tgw-07718388470c8a2c1					

Add route

Cancel

Preview

Save changes

Step 14: Configure Security Groups

Allow

SSH

Type Source

SSH Other VPC CIDR

Example

For EC2-A

Allow

20.0.0.0/16

For EC2-B

Allow

10.0.0.0/16

Allow ICMP if you want to test ping.

The screenshot shows the 'Edit inbound rules' page in the AWS Management Console. The breadcrumb trail at the top is 'EC2 > Security Groups > sg-0eb70f559a27ad5c3 > Edit inbound rules'. The page title is 'Edit inbound rules' with an 'Info' link. Below the title is a note: 'Inbound rules control the incoming traffic that's allowed to reach the instance.' The main content area is titled 'Inbound rules' and contains a table with columns: 'Security group rule ID', 'Type', 'Protocol', 'Port range', 'Source', and 'Description - optional'. There are four rules listed:

Security group rule ID	Type	Protocol	Port range	Source	Description - optional	Actions
sgr-0eb3aceda237daa9e	SSH	TCP	22	Custom	0.0.0.0/0	Delete
-	All ICMP - IPv4	ICMP	All	Anywh...	0.0.0.0/0	Delete
-	SSH	TCP	22	Custom	20.0.0.0/16	Delete
-	All ICMP - IPv4	ICMP	All	Custom	20.0.0.0/16	Delete

At the bottom left of the table is an 'Add rule' button. At the bottom right of the page are three buttons: 'Cancel', 'Preview changes', and 'Save rules'.

Step 15: Test Transit Gateway

SSH into EC2-A

Ping EC2-B Private IP

ping 20.x.x.x

Fill values:

Name tag

S3-Endpoint

Service category

Select:

AWS services

Services

Search:

S3

Select:

com.amazonaws.us-east-1.s3

Example:

Type: Gateway

Service name:

com.amazonaws.us-east-1.s3

VPC selection

Choose:

vpc-0e14a41c99469658d

Example:

VPC ID:

Shared-VPC

3. Route Tables

Select:

Route tables

Choose your **private route table**.

Example:

Private-RT-Shared

rtb-xxxxxxx

Do NOT choose public route table.

Because:

Private EC2 → S3 should use Endpoint.

4. Policy

Select:

Full Access

This creates:

```
{  
  "Effect": "Allow",  
  "Principal": "*",  
  "Action": "s3:*",  
  "Resource": "*" }  
}
```

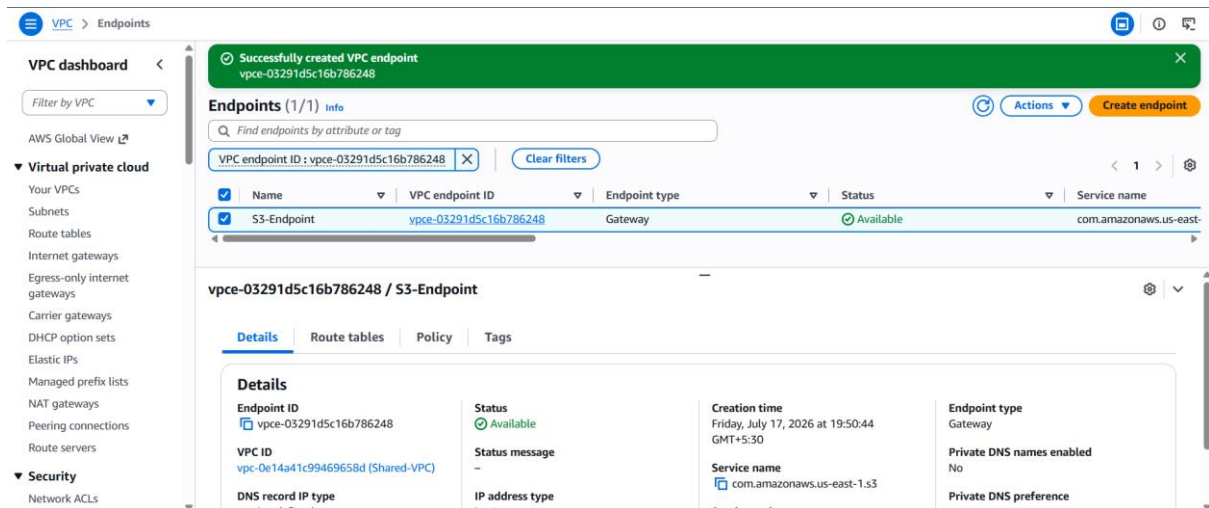
5. Create Endpoint

Click:

Create endpoint

Wait until:

Status: Available



Step 17: Verify Route Table

Now AWS automatically adds a route.

Go to:

VPC

→ Route Tables

→ Private Route Table

Open:

Routes

You should see:

Example:

Destination Target

10.0.0.0/16 local

pl-63a5400a vpce-xxxxxxx

Where:

pl-63a5400a

is the S3 prefix list.

Example:

Destination:

pl-63a5400a

Target:

vpce-0123456789

This means:

EC2 → VPC Endpoint → S3

No Internet Gateway required.

Step 18: Launch EC2 in Shared Services VPC

Now create an EC2 instance.

Go:

EC2

→ Instances

→ Launch Instance

EC2 > Instances

Dashboard

AWS Global View

Events

Instances

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

Capacity Manager

Instances (1/3) Info

Saved filter sets

Choose filter set

Find Instance by attribute or tag (case-sensitive)

Instance state = running

Clear filters

Connect

Instance state

Actions

Launch instances

☒	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	Put
☒	EC2-Share	i-0eab58aec4581c3c2	Running	t3.micro	3/3 checks passed	us-east-1c	-	-
☐	EC2-A	i-045730f96d46e6b81	Running	t3.micro	3/3 checks passed	us-east-1a	ec2-44-198-175-187.co...	44

i-0eab58aec4581c3c2 (EC2-Share)

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

Instance summary Info

Instance ID

i-0eab58aec4581c3c2

IPv6 address

-

Hostname type

IP name: ip-30-0-1-151.ec2.internal

Answer private resource DNS name

-

Public IPv4 address

-

Instance state

Running

Private IP DNS name (IPv4 only)

ip-30-0-1-151.ec2.internal

Instance type

t3.micro

Private IPv4 addresses

30.0.1.151

Public DNS

-

Elastic IP addresses

-

Verify with command: `aws s3 ls`

```
brajk@RajKumari MINGW64 ~/Downloads
$ ssh -i linux-testing.pem ec2-user@98.92.34.148

#_
~\#####_ Amazon Linux 2023
~~~\#####\
~~~\###|
~~~\#/ https://aws.amazon.com/linux/amazon-linux-2023
~~~~V~'-'>
~~~~_.-.-_-/_
~/m/'-_-/_

Last login: Thu Jul 16 17:31:12 2026 from 103.248.210.235
[ec2-user@ip-10-0-1-182 ~]$ aws --version
aws-cli/2.33.15 Python/3.9.25 Linux/6.18.36-69.138.amzn2023.x86_64 source/x86_64.amzn.2023
[ec2-user@ip-10-0-1-182 ~]$ aws s3 ls
2026-07-15 18:35:16 vpc-flowlogs-bucket-raji
[ec2-user@ip-10-0-1-182 ~]$
```